Supporting Information

ChemIP: An Open-Source Web Platform for Unified Chemical Safety Data Retrieval across Korean and International Regulatory Sources

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S1. Platform Installation and Configuration

Prerequisites

- Python 3.11+ with pip
- Node.js 18+ with npm
- (Optional) Ollama for local LLM features

Step 1: Clone and install dependencies

```
git clone https://github.com/yuyongkim/chemIP.git
cd chemIP

# Backend
pip install -r requirements.txt

# Frontend
cd frontend && npm install && cd ..
```

Step 2: Configure environment variables

Copy `.env.example` to `.env` and provide the following API keys:

Variable	Source	Required
`KOSHA_SERVICE_KEY_DECODED`	Korea Occupational Safety and Health Agency	Yes
`KIPRIS_API_KEY`	Korea Intellectual Property Rights Information Service	Yes
`KOTRA_API_KEY_DECODED`	Korea Trade-Investment Promotion Agency	Yes
`DRUG_API_KEY_DECODED`	Ministry of Food and Drug Safety (MFDS)	Yes
`PUBMED_API_KEY`	NCBI E-utilities	No
`NAVER_CLIENT_ID` / `NAVER_CLIENT_SECRET`	Naver Developers (news fallback)	No

All Korean government API keys are available at no cost through the respective open data portals (data.go.kr for KOSHA/MFDS, kipris.or.kr for KIPRIS, openapi.kotra.or.kr for KOTRA).

Step 3: Start services

```
# Backend (FastAPI)
uvicorn backend.main:app --host 127.0.0.1 --port 7010

# Frontend (Next.js)
cd frontend && npm run dev
```

Access points after startup:

Service	URL
Web interface	http://localhost:7000
API server	http://localhost:7010
Interactive API docs (Swagger)	http://localhost:7010/docs
Health check	http://localhost:7010/health

Step 4: Production deployment (optional)

For persistent operation, a PM2 process manager configuration is provided:

```
# Windows
start_all.bat

# Linux/macOS
bash start_all.sh
```

S2. API Endpoint Reference

ChemIP exposes a RESTful API organized into seven route modules. All endpoints are prefixed with `/api/`.

S2.1 Chemical Substance Endpoints (`/api/chemicals`)

Method	Endpoint	Description
GET	`/`	Full-text search by chemical name or CAS number (paginated)
GET	`/autocomplete`	Typeahead suggestions for chemical names
GET	`/{chem_id}`	Retrieve complete MSDS data (all 16 sections) with GHS classification
GET	`/{chem_id}/drugs`	Linked pharmaceutical data for a chemical (MFDS/OpenFDA/PubMed)

When a chemical detail is requested and MSDS sections are not cached locally, the system fetches all 16 sections from the KOSHA API in parallel and stores them for subsequent requests.

S2.2 Patent Endpoints (`/api/patents`)

Method	Endpoint	Description
GET	`/`	Search Korean patents via KIPRIS keyword query
GET	`/kipris/{application_number}`	Retrieve detailed KIPRIS patent record
GET	`/uspto/{chem_id}`	Search local USPTO patent index by chemical identifier
GET	`/global/{chem_id}`	Search global patent index (USPTO/EPO/WIPO/CN/JP/KR)

The global patent index contains 350M+ patent records stored in a local SQLite database (134.5 GB), enabling sub-second full-text search without external API calls.

S2.3 Drug and Pharmaceutical Endpoints (`/api/drugs`)

Method	Endpoint	Description
GET	`/approval`	MFDS drug approval database search
GET	`/easy`	MFDS consumer medication information
GET	`/openfda`	US drug labels by brand, generic, or substance name
GET	`/pubmed`	Biomedical literature search (multi-query: base + toxicology + safety)
GET	`/unified`	Parallel search across all three sources (MFDS + OpenFDA + PubMed)

S2.4 Trade Intelligence Endpoints (`/api/trade`)

Method	Endpoint	Description
GET	`/news`	KOTRA overseas market news
GET	`/strategy`	Market entry strategy reports by country
GET	`/prices`	Product price trend information
GET	`/fraud`	Trade fraud case database
GET	`/import-regulations`	Import restrictions by country
GET	`/naver-news`	Korean news search (Naver, fallback source)

Additional endpoints exist for country information, enterprise success cases, and tourism data.

S2.5 Safety Guide Endpoints (`/api/guides`)

Method	Endpoint	Description
GET	`/status`	KOSHA guide dataset availability and statistics
GET	`/search`	Full-text search across KOSHA safety guides
GET	`/{guide_no}`	Retrieve detailed KOSHA safety guide by ID
GET	`/recommend/{chem_id}`	Recommended safety guides for a chemical (multi-criteria matching)

S2.6 AI Analysis Endpoints (`/api/ai`)

These endpoints require a local Ollama LLM server (default model: qwen3:8b).

Method	Endpoint	Description
POST	`/analyze`	Multi-source safety analysis report (MSDS + patents + guides)
POST	`/summarize`	Comprehensive safety summary across all data sources
POST	`/drug-analysis`	Drug-chemical relationship analysis
POST	`/recommend`	Related chemical/keyword recommendations
POST	`/ask`	Natural language Q&A with full data context
GET	`/llm-status`	Check LLM server availability

S2.7 Regulatory Classification Endpoints (`/api/regulations`)

Source	Endpoints	Data Provided
ECHA (EU)	`/echa/search`, `/echa/clp/{id}`, `/echa/labelling/{id}`	REACH registration, CLP classification, harmonised labelling
EPA CompTox (US)	`/comptox/search`, `/comptox/hazard/{dtxsid}`	ToxValDB hazard data, cancer classification, functional use
NIOSH (US)	`/niosh/search`, `/niosh/cas/{cas}`, `/niosh/exposure/{cas}`	REL/PEL/IDLH exposure limits, PPE recommendations, carcinogen list
KISCHEM (KR)	`/kischem/search`, `/kischem/cas/{cas}`	Exposure symptoms, first-aid procedures
NCIS (KR)	`/ncis/cas/{cas}`, `/ncis/search`	Korean environmental substance classification
Aggregated	`/search`	Cross-source search (all five sources in parallel)

Total: 80+ API endpoints across all modules.

S3. Database Schema

ChemIP uses three local SQLite databases:

S3.1 terminology.db (52.6 MB) — Primary chemical and MSDS database

Table: `chemical_terms` (117,744 rows)

Column	Type	Description
id	INTEGER PK	Internal row ID
name	TEXT	Chemical name (Korean or mixed)
cas_no	TEXT	CAS registry number
name_en	TEXT	English name (when available)
source	TEXT	Data source: KOSHA, ECHA, KREACH, KISCHEM, PUBCHEM
external_id	TEXT	External source identifier

Source distribution: ECHA 80,534 (68.4%), KREACH 21,804 (18.5%), KOSHA 13,448 (11.4%), KISCHEM 1,954 (1.7%), PubChem 4 (<0.1%).

Table: `msds_details` (550,161 rows)

Column	Type	Description
chem_id	TEXT	Chemical identifier
section_no	INTEGER	MSDS section number (1-16)
xml_data	TEXT	XML-formatted section content from KOSHA API

Table: `msds_english` (56,176 rows)

Column	Type	Description
chem_id	TEXT	Chemical identifier
cas_no	TEXT	CAS registry number
name_en	TEXT	English chemical name
pubchem_cid	INTEGER	PubChem compound ID
signal_word	TEXT	GHS signal word (Danger/Warning)
ghs_classification	TEXT	JSON array of GHS classifications
hazard_statements	TEXT	JSON array of H-statements
precautionary_statements	TEXT	JSON array of P-statements
pictograms	TEXT	JSON array of GHS pictogram codes

Table: `chemical_terms_fts` — FTS5 virtual table for full-text search over name, cas_no, name_en.

S3.2 global_patent_index.db (134.5 GB) — Global patent full-text index

Contains patent records from USPTO, EPO, WIPO, CN, JP, and KR patent offices with chemical name matching via FTS5 indexing.

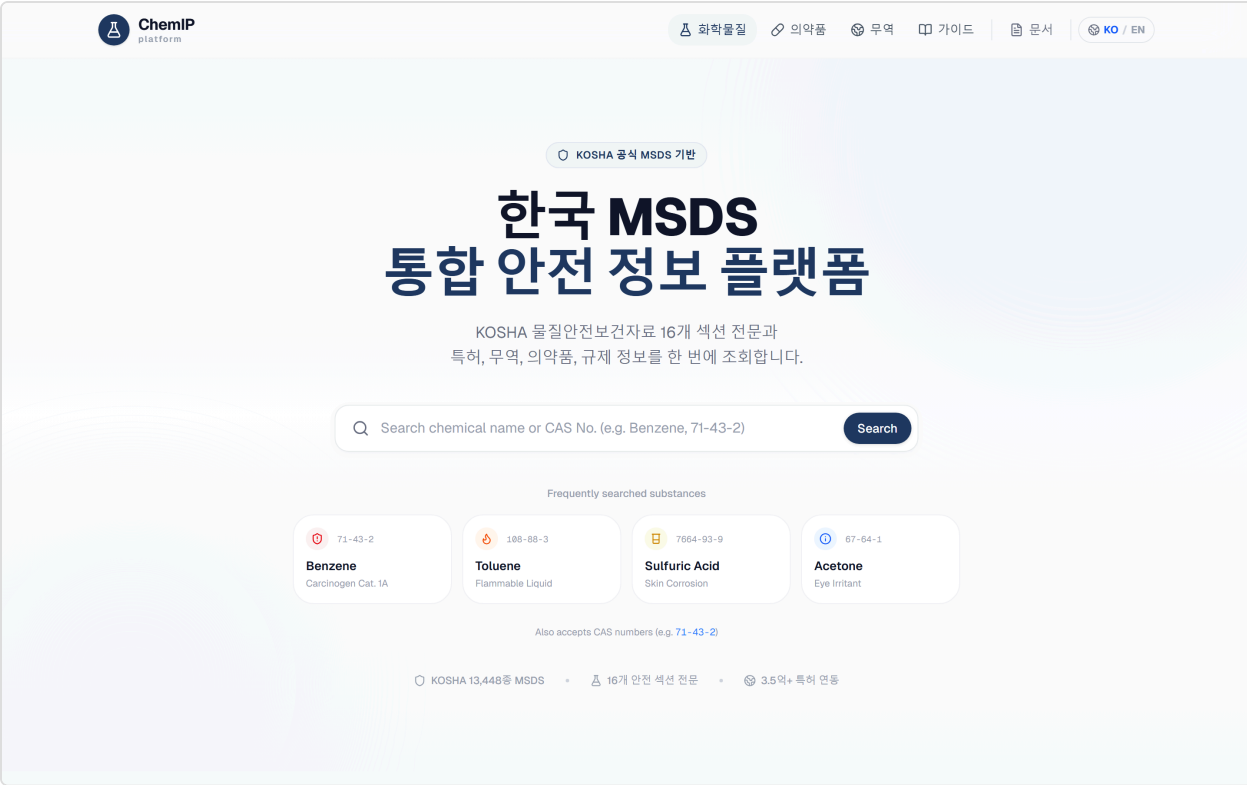
S3.3 uspto_index.db (5.8 GB) — US patent index

Dedicated USPTO patent index for faster US-specific patent queries.

S4. User Interface Screenshots

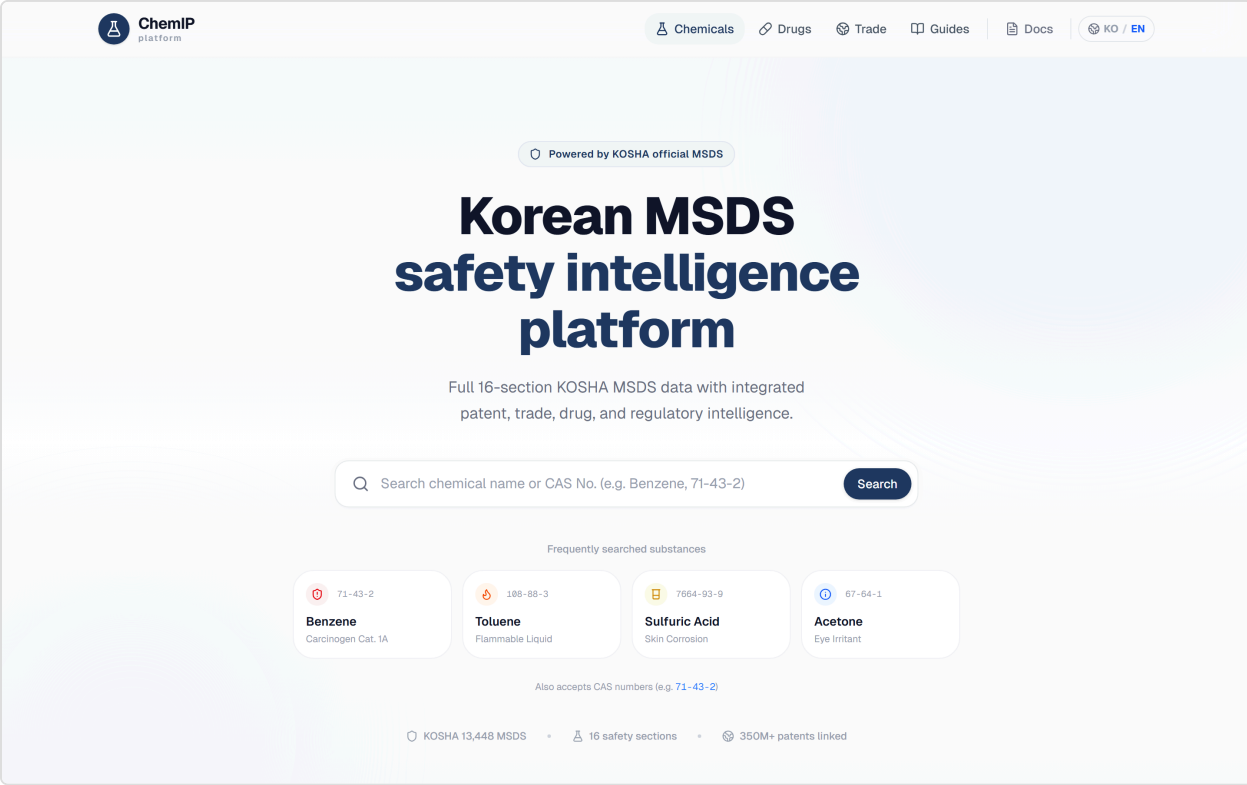
The platform supports bilingual (Korean/English) operation. Representative screenshots from the live deployment (https://chemip.yule.pics) are shown below.

Figure S1. Home page — Korean interface



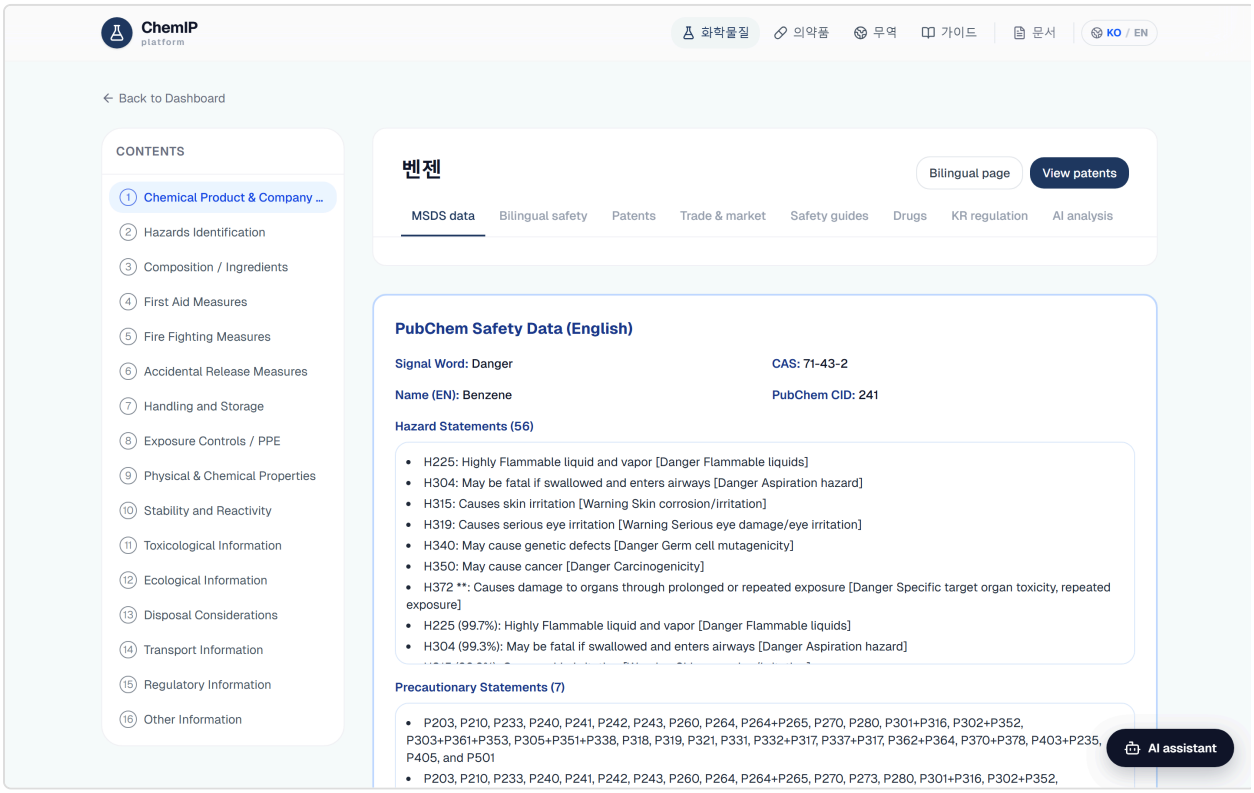
The Korean home page displays "한국 MSDS 통합 안전 정보 플랫폼" (Korean MSDS Integrated Safety Information Platform) with KOSHA as the primary data source. Statistics show 13,448 KOSHA MSDS entries with full 16-section coverage and 350M+ linked patents.

Figure S2. Home page — English interface



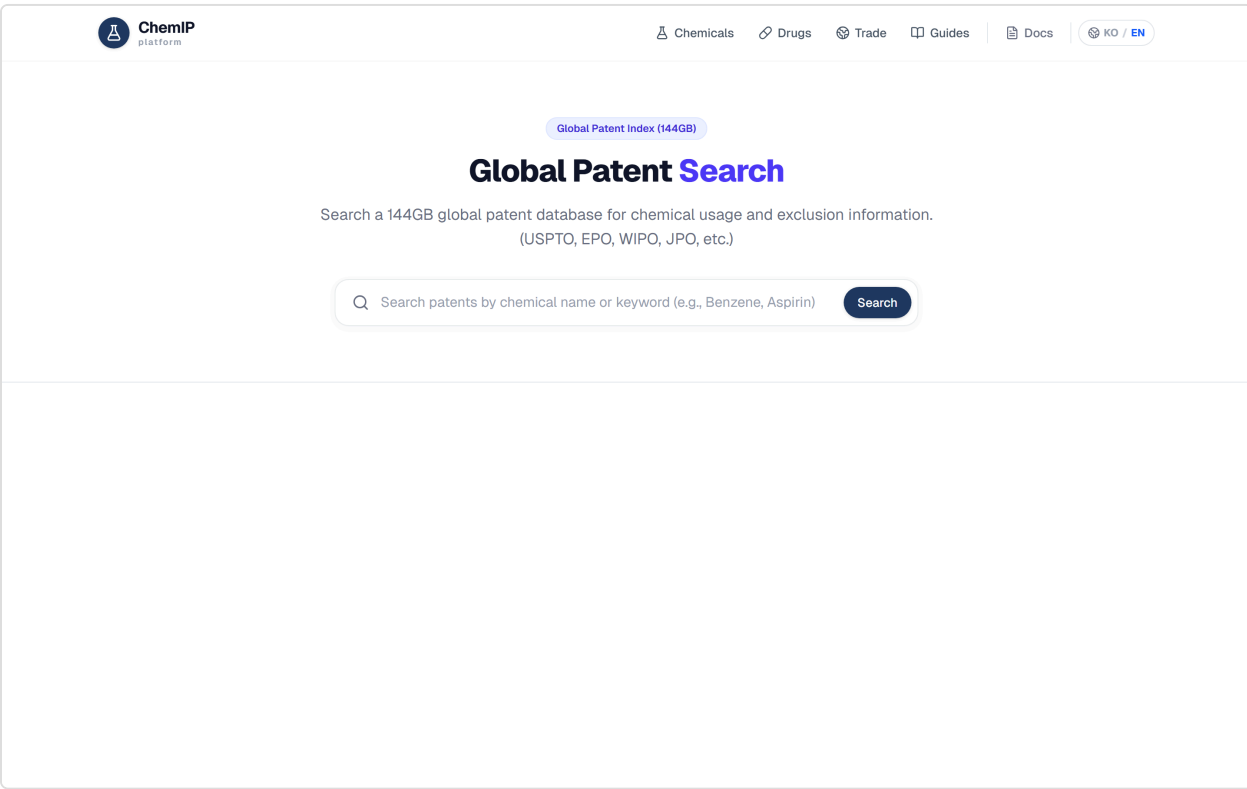
The same home page in English mode, demonstrating the bilingual toggle in the navigation bar. All UI labels, descriptions, and navigation items are translated.

Figure S3. Chemical detail page (Benzene, CAS 71-43-2)



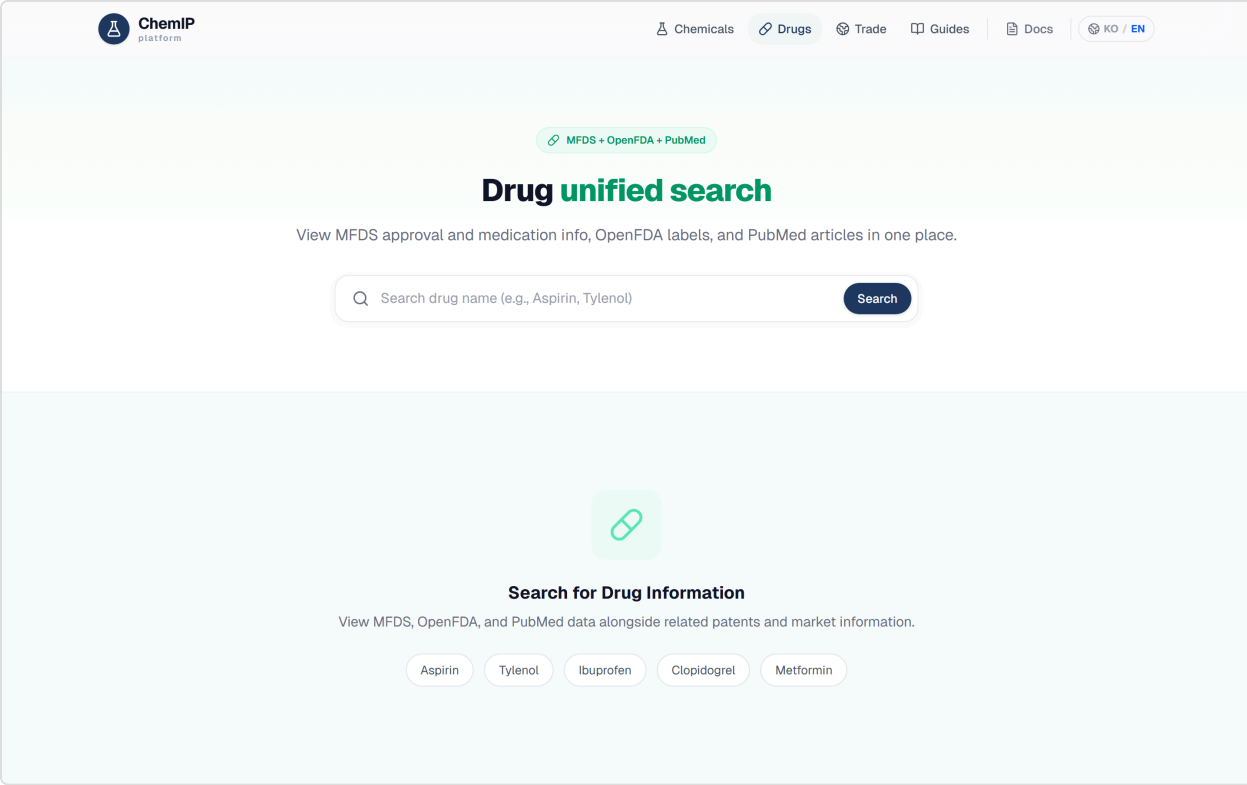
The chemical detail page for benzene showing: (left) table-of-contents sidebar for MSDS sections 1–16; (right) PubChem English safety data with GHS signal word, hazard statements (H225, H304, H315, H319, H340, H350, H372), and precautionary statements. Tabs provide access to bilingual safety view, patents, trade data, safety guides, drugs, Korean regulation, and AI analysis.

Figure S4. Patent search interface



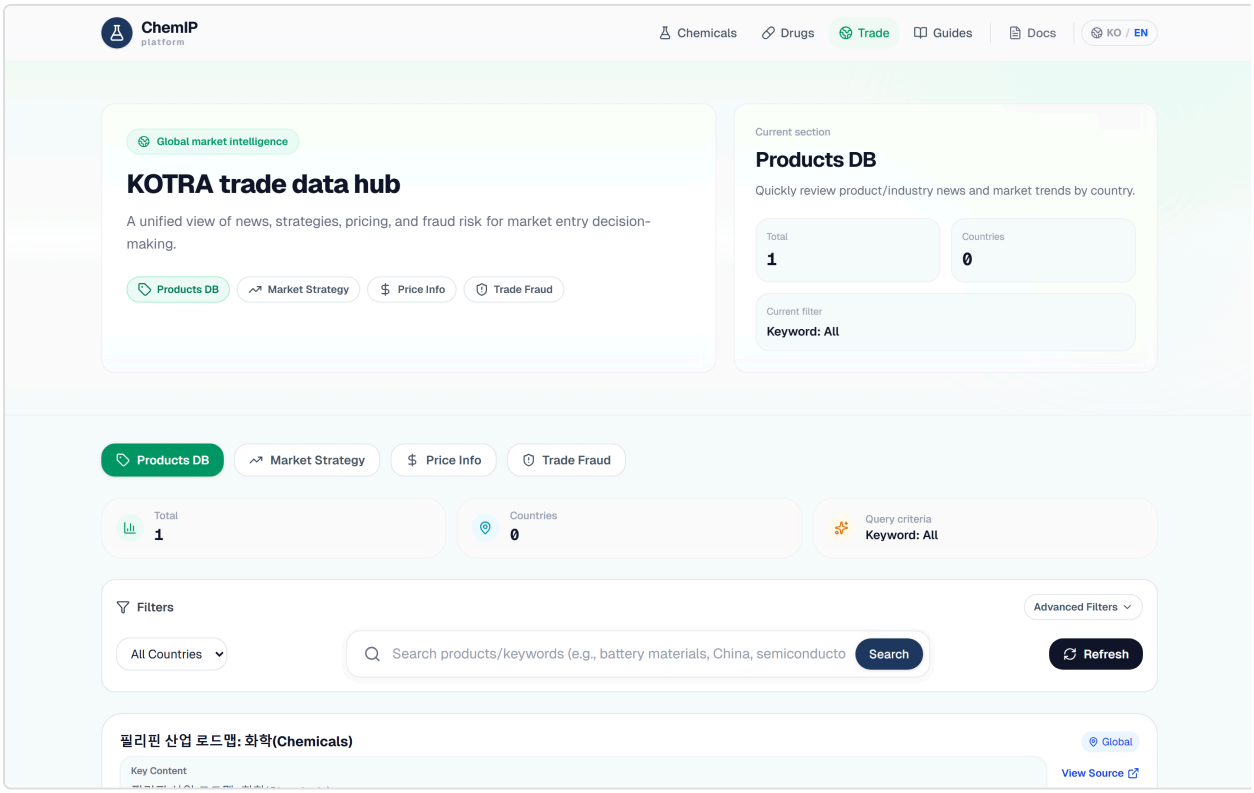
The patent search page providing access to KIPRIS (Korean patents) and the local global patent index (USPTO/EPO/WIPO/CN/JP/KR, 350M+ records).

Figure S5. Drug cross-reference panel



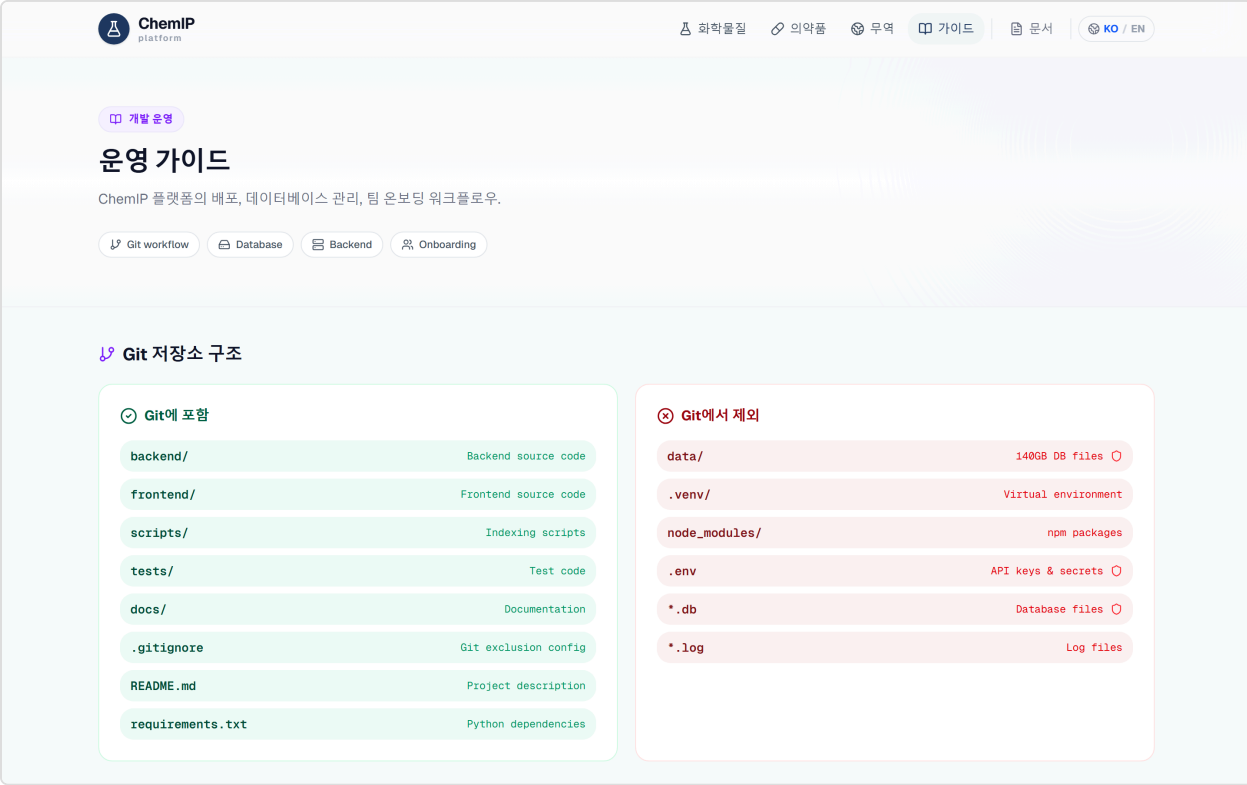
The unified drug search interface combining MFDS (Korean drug approvals and consumer medication info), OpenFDA (US drug labels), and PubMed (biomedical literature) in a single query.

Figure S6. Trade intelligence dashboard



The KOTRA trade data hub showing product database, market strategy, price information, and trade fraud tabs. The interface displays result counts, country coverage, and current filter criteria.

Figure S7. Operations guide page



The operations guide page providing Git workflow documentation, database sharing methods, backend operation commands, and team onboarding checklists for platform deployment.

S5. Deployment Modes

ChemIP supports three deployment configurations to accommodate different resource constraints:

Mode	Components	Database	Storage	Use Case
Minimal	Backend + Frontend	terminology.db only (53 MB)	~200 MB	MSDS lookup and drug search
Standard	+ KOSHA Guides + Ollama LLM	+ guide dataset (~500 MB)	~4 GB	Full safety analysis with AI
Full	+ Global patent index	+ patent DBs (140 GB)	~145 GB	Complete platform with patent search

All modes support the same API interface; unavailable features return graceful error messages rather than failing silently, consistent with the adapter-based graceful degradation design described in the main text.

This Supporting Information accompanies the manuscript "ChemIP: An Open-Source Web Platform for Unified Chemical Safety Data Retrieval across Korean and International Regulatory Sources" submitted to ACS Chemical Health & Safety.